21-242 Recitation 12

Matrices — Q3. [T] [ ] =[T( )]

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $[T]_{\mathcal{CB}}[\vec{v}]_{\mathcal{B}} = [T(\vec{v})]_{\mathcal{C}}$.

Course-level solution

Write $\vec{v} = \sum_{j=1}^{n} x_j \vec{b}_j$, so $[\vec{v}]_{\mathcal{B}} = \vec{x} = (x_1, \ldots, x_n)^T$. Then $T(\vec{v}) = \sum_{j=1}^{n} x_j T(\vec{b}_j)$, hence

$$[T(\vec{v})]_{\mathcal{C}} = \sum_{j=1}^{n} x_j [T(\vec{b}_j)]_{\mathcal{C}} = [T]_{\mathcal{CB}}\, \vec{x} = [T]_{\mathcal{CB}}\, [\vec{v}]_{\mathcal{B}},$$

because multiplying a matrix by $\vec{x}$ is the corresponding linear combination of its columns.